L-bracket: HOLLOW DESIGN ACCEPTED

```
cshell design accepted: J = +0.0687 (optimisation grid), +0.0700 (shifted grid) > ln(1 + g_min) = 0.0198,
feasible, export mass-matched and watertight

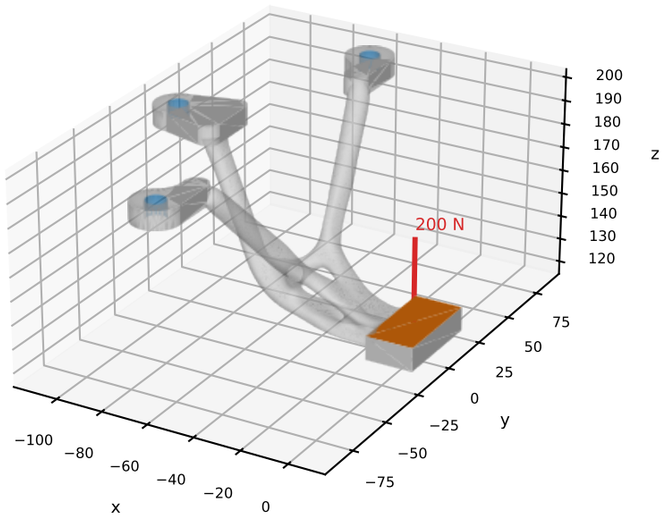
mass      target 43.63 g   final 43.63 g   error 4.5725e-06
optimiser grid  J 0.06868 (weighted)  ln K/K0 0.08692  K/K0 1.091  P/P0 1.052  governing strength  feasible True   (accept: J
    > ln 1.02 = 0.0198, K/K0 >= 0.85, P/P0 >= 0.85)
shifted grid    J 0.06998 (weighted)  ln K/K0 0.08692  K/K0 1.091  P/P0 1.054  governing strength  feasible True   (accept: J
    > ln 1.02 = 0.0198, K/K0 >= 0.85, P/P0 >= 0.85)

export    h 0.2 mm  faces 1562200  watertight True  volume 4.195e+04 mm3  material bodies 1  shells 5  enclosed voids 4
    escape holes 0
wall [mm] min 1.469  p05 1.549  median 5.303  max 42.22
closed_area_frac   0.4815
hollow_area_frac   0.2652

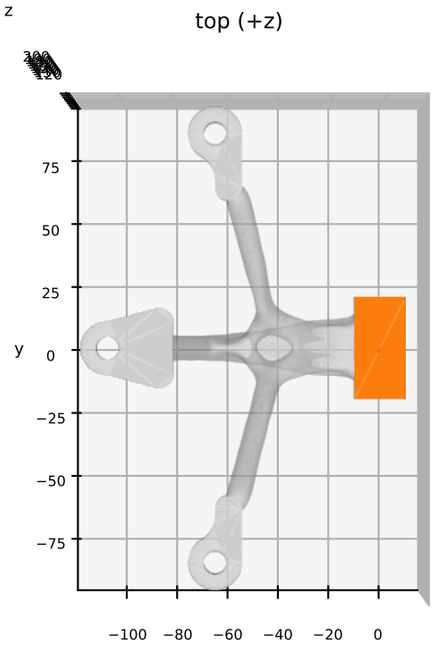
optimiser: {'method': 'mma', 'starts': ['solid'], 'max_iter': [60, 80, 40], 'move': 0.1, 'tol_x': 0.005, 'tol_f': 0.0001,
    'mass_tol': 0.0001, 'sliver_weight': [0.0, [0.0, 0.05], 0.1], 'seed': 0}
grid: {'phases': [{'h_e': 0.8, 'h_g': 0.2}, {'h_e': 0.4, 'h_g': 0.1}], 'export_h_mm': 0.05, 'rho_min': 1e-06,
    'tessellation_mm': 0.02, 'active_margin_mm': 0.5, 'geometry_device': 'auto', 'geometry_offload': True}
growth: {'enabled': True, 'phase': 0, 'tol': 0.001, 'patience': None, 'quantiles': [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8,
    0.9, 1.0], 'step_tol': 0.0001, 'max_iters_per_step': 25, 'min_iters_per_step': 2, 'samples': 16}
objective: {'type': 'weighted', 'constraints': [], 'constraint_scale': 0.01, 'feasibility_tol': 0.001, 'w_stiffness': 0.5,
    'w_capacity': 0.5, 'min_gain': 0.02, 'floor_stiffness': 0.85, 'floor_capacity': 0.85, 'floors_as_constraints': True}
export cfg: {'h_mm': 0.2, 'device': 'gpu', 'closed_thr': 0.9, 'escape_holes': False}
```

L-bracket: supports (blue, all DOF fixed; 3 patches) and loads (orange faces; 1 red arrow, one per loaded patch at its area centroid, length and label = its share of the total 200 N). Faces read from bc_faces.npz (as analysed).

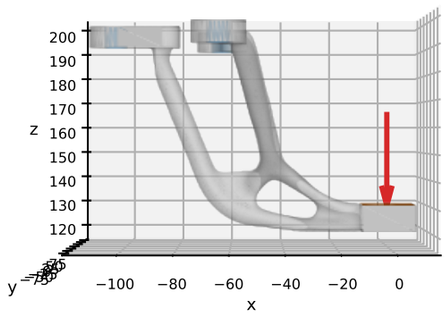
isometric



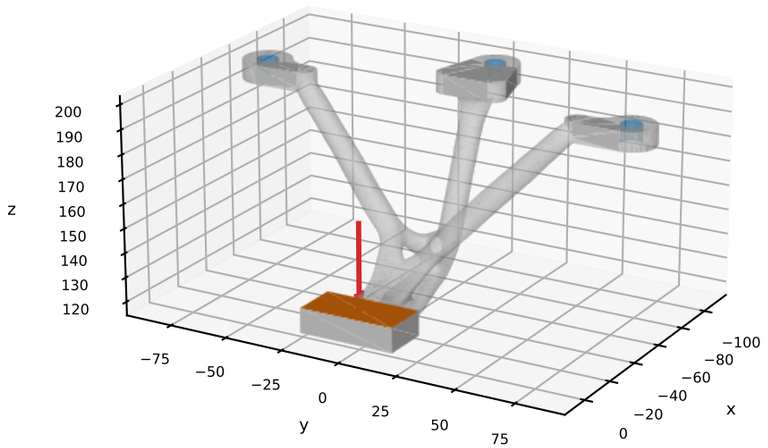
top (+z)



front (-y)

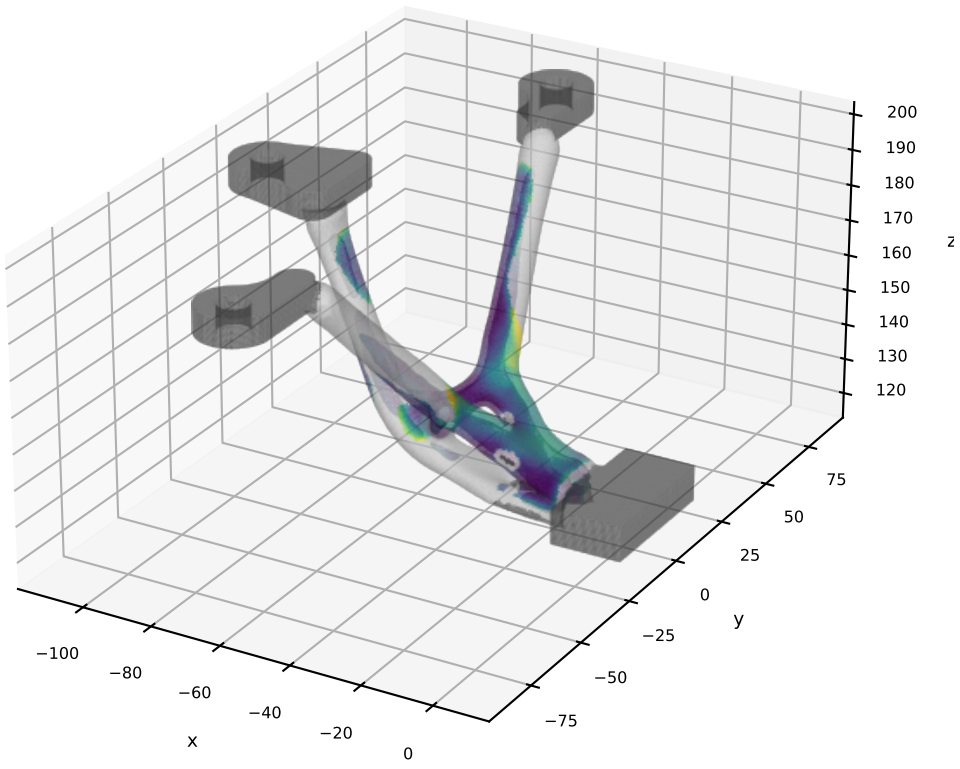


rear isometric

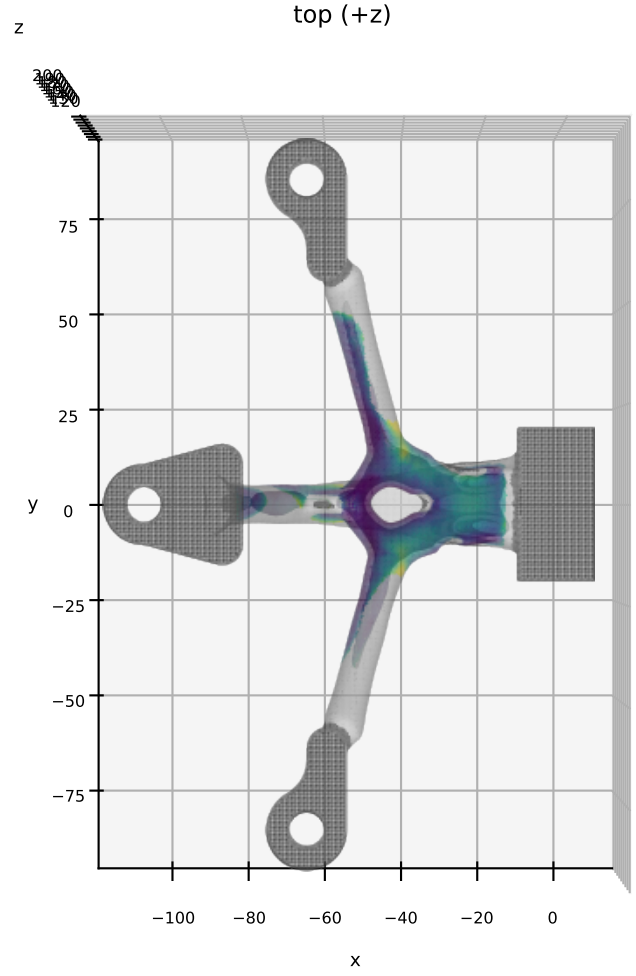


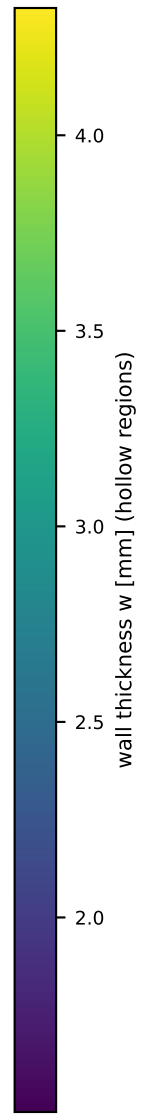
L-bracket: final design (STL 1,562,200 faces, shown decimated to 159,126; wall thickness from fields.vtp)

isometric

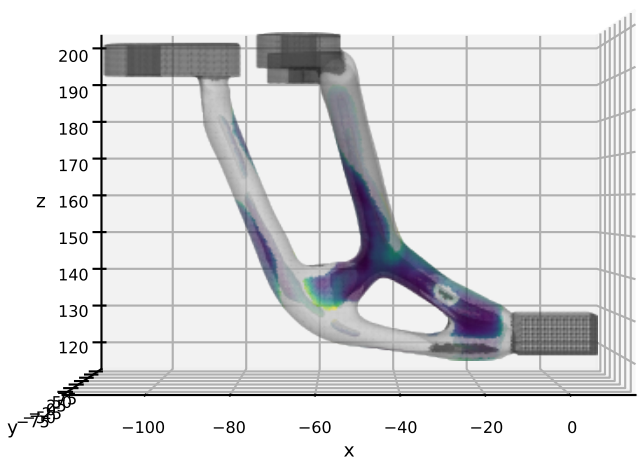


top (+z)

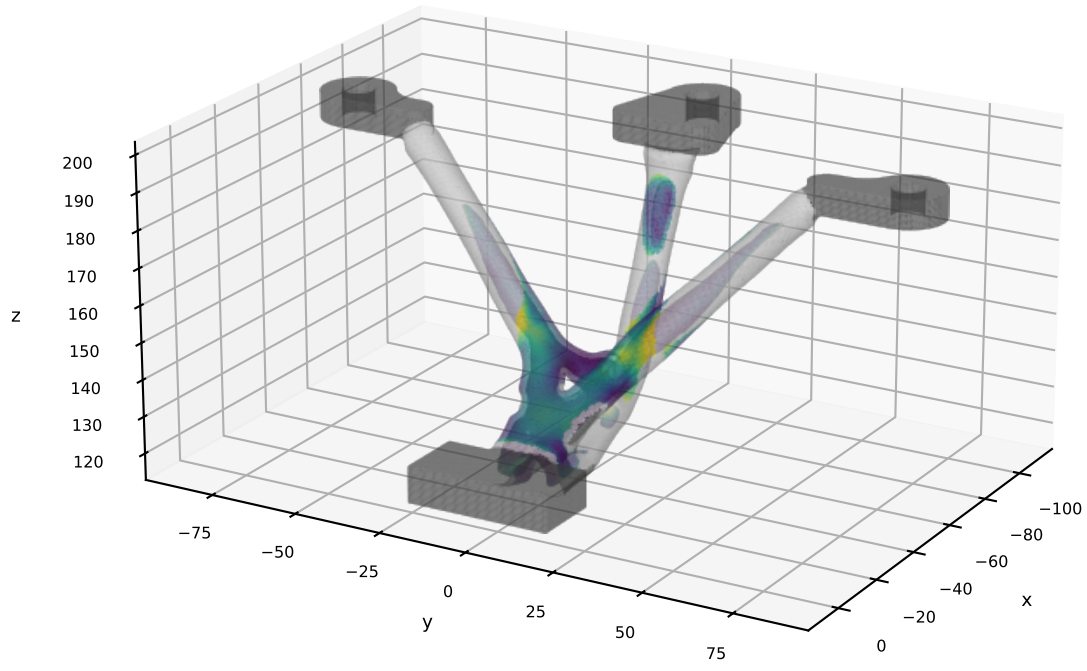




front (-y)

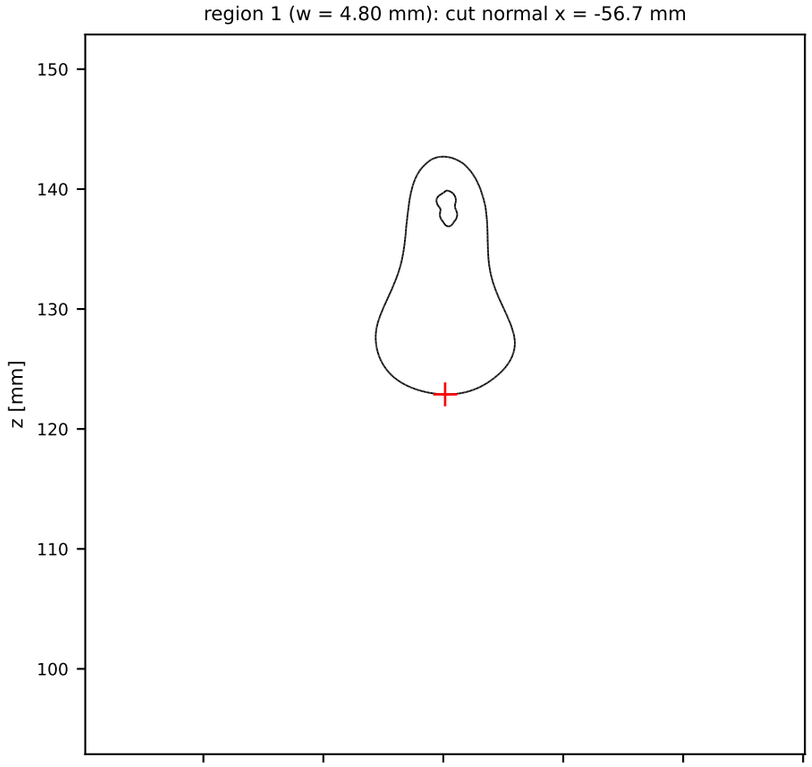


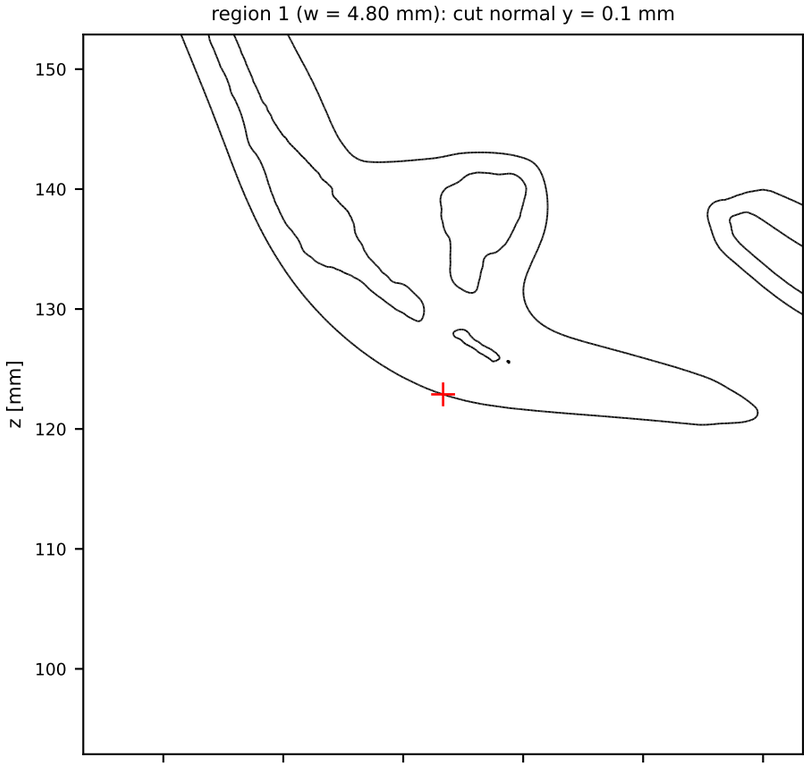
rear isometric

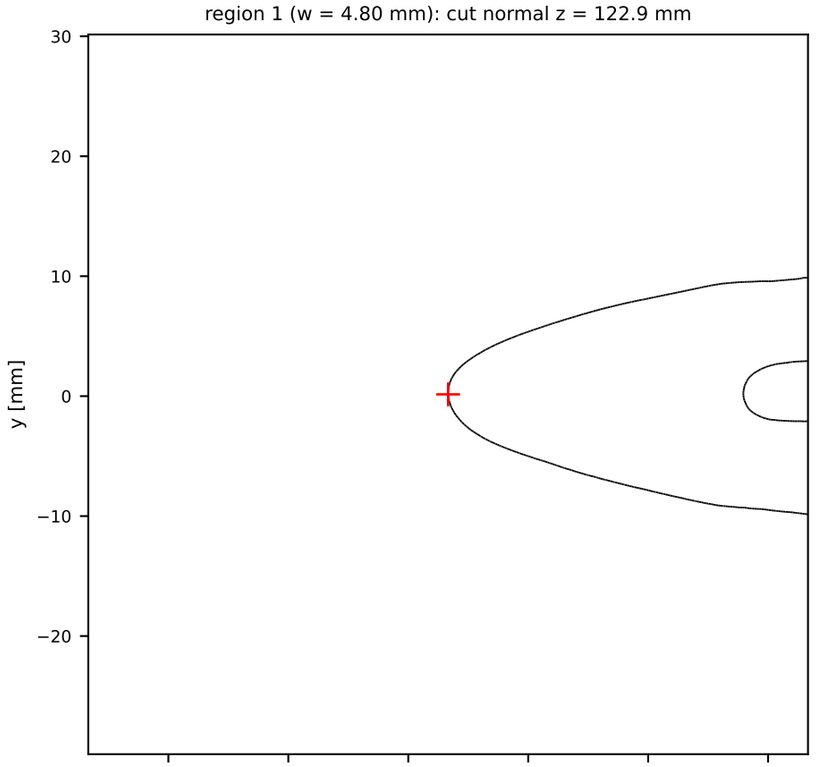


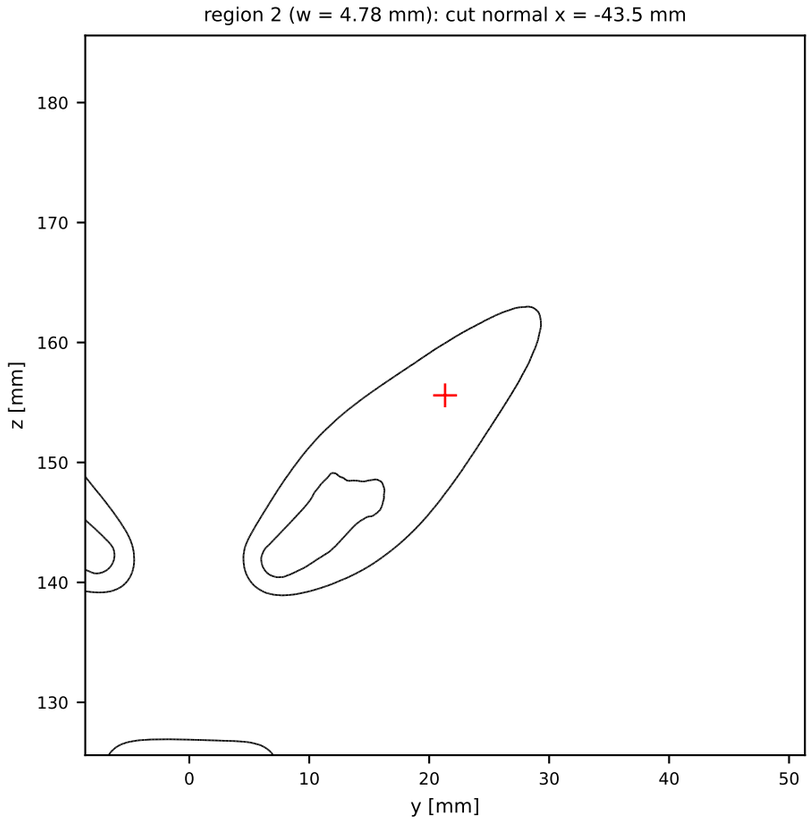
light grey = solid (closed) regions; dark grey = mounts; the colourbar covers the hollow regions only

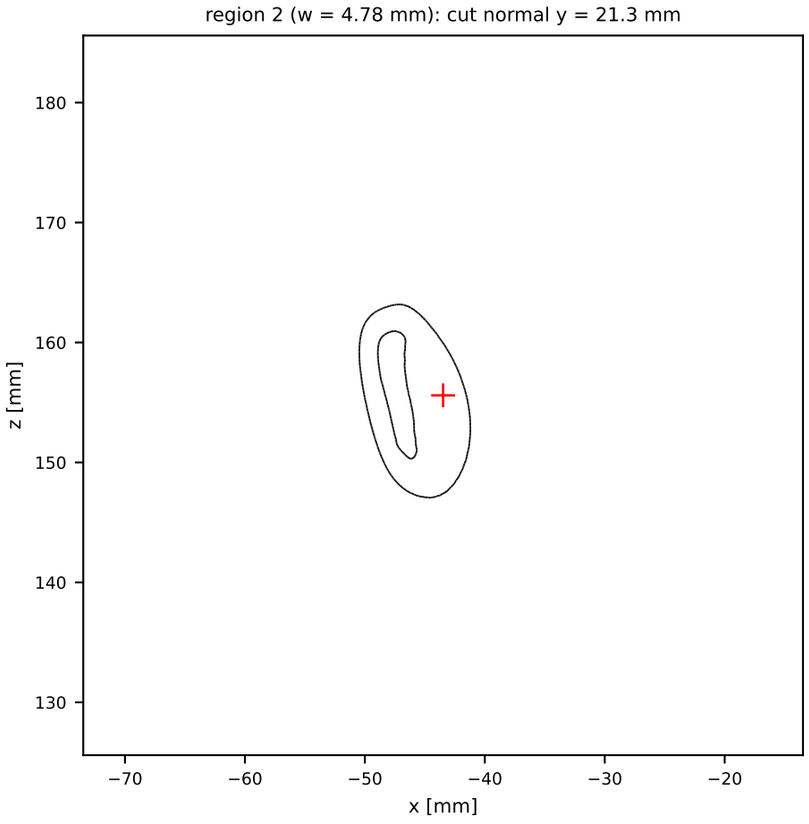
L-bracket: sections of the final design through the two thickest-wall regions (red +); +-30 mm window

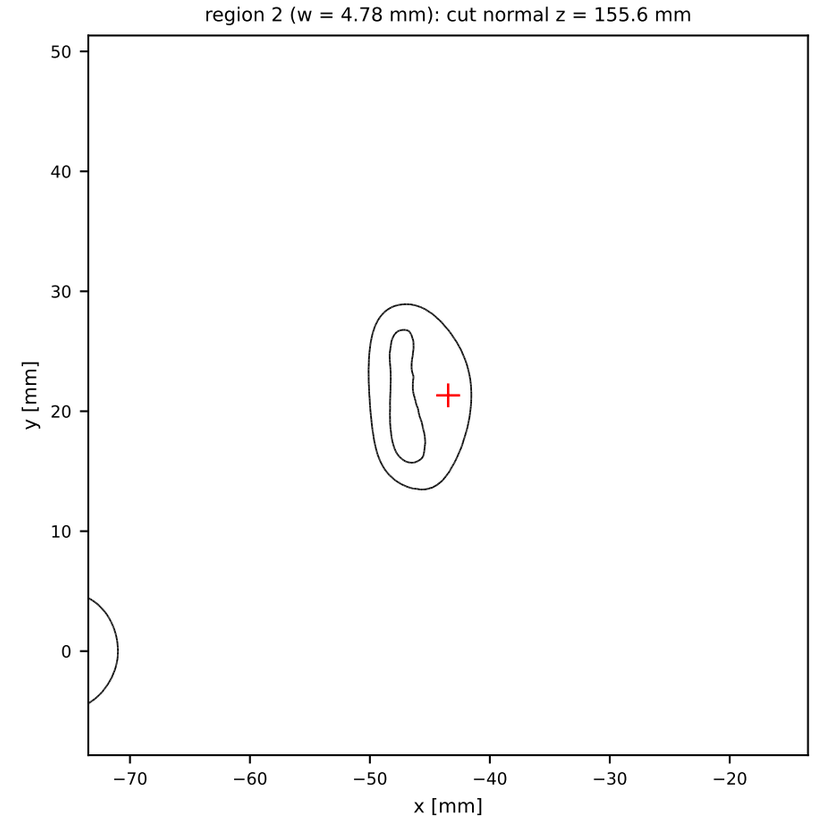






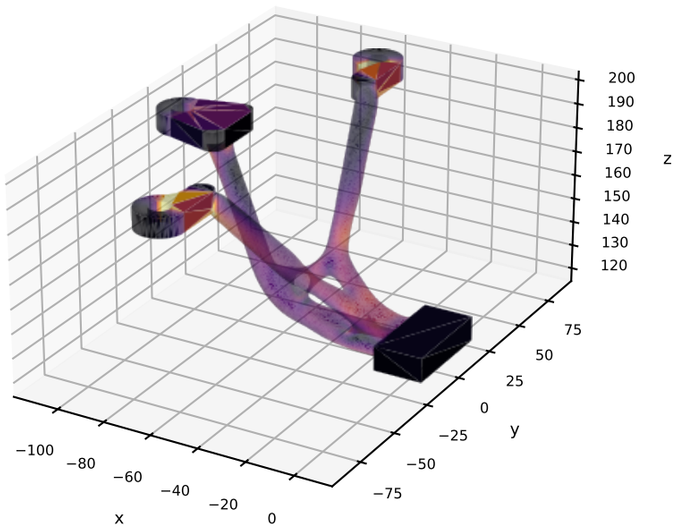




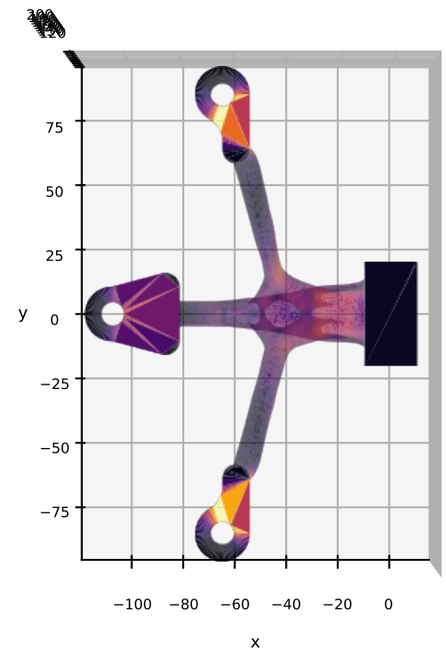


L-bracket: von Mises stress, original solid vs final design (peak solid 42.6, final 41.3 MPa; values from fields.vtp, nearest point)

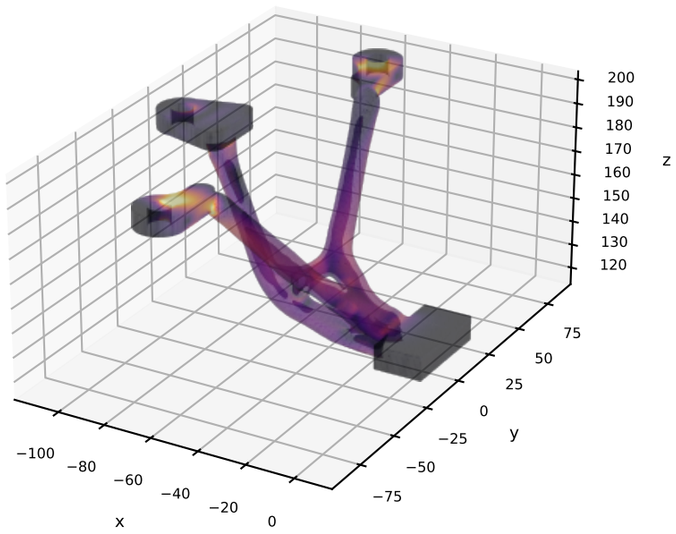
solid: isometric



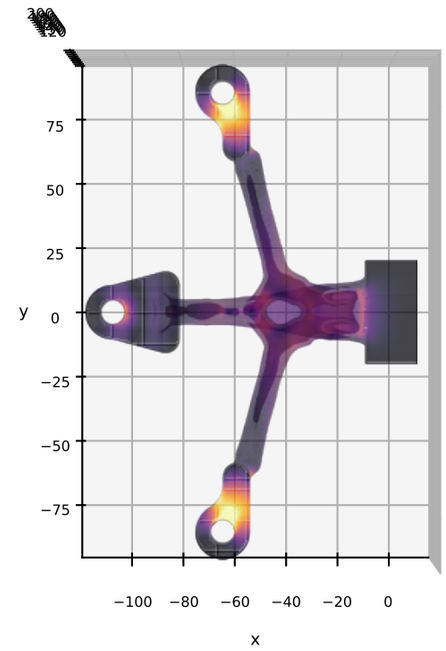
solid: top (+z)

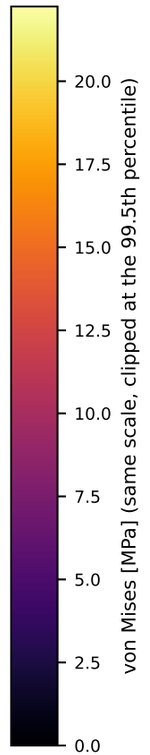


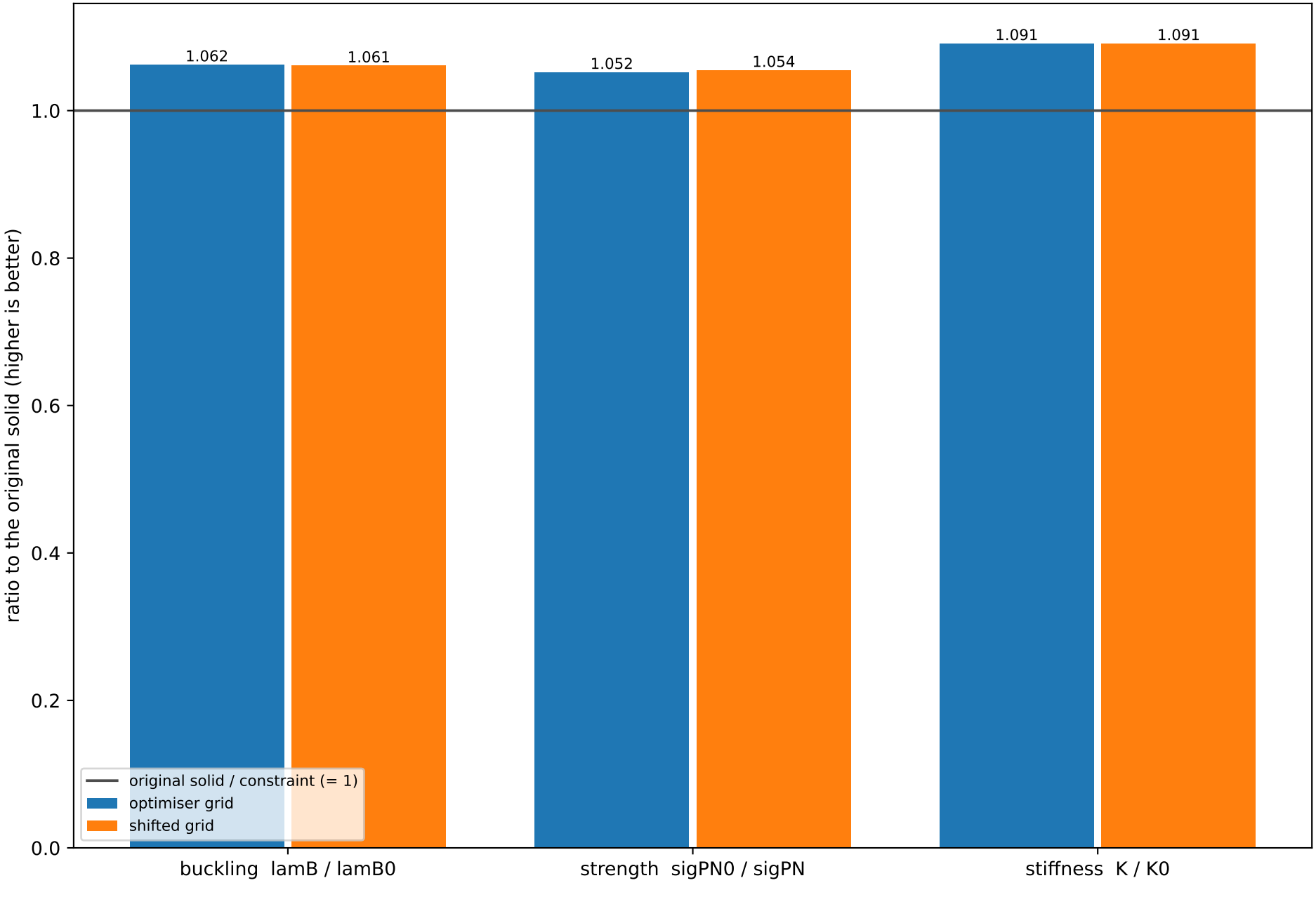
final design: isometric

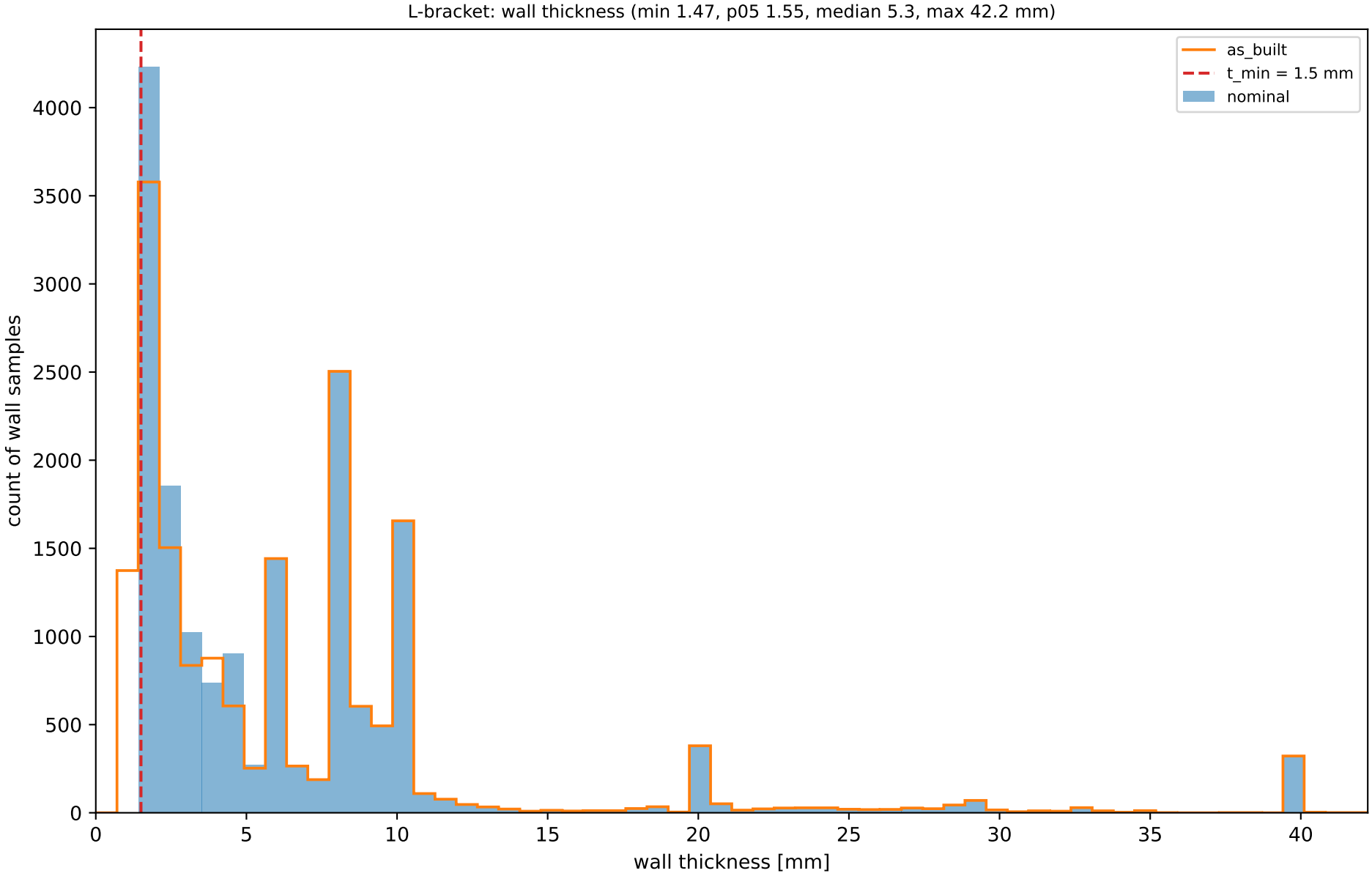


final design: top (+z)



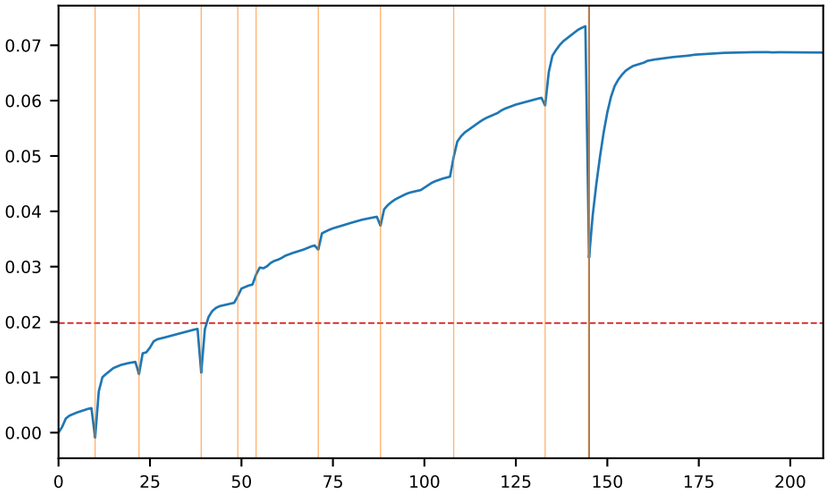



L-bracket: capacity by mode against the solid's constraints. The buckling and strength constraints require both bars >= 1.
ratio to the original solid (higher is better)
0.0
0.2
0.4
0.6
0.8
1.0
1.062
1.061
1.052
1.054
1.091
1.091
original solid / constraint (= 1)
optimiser grid
shifted grid
buckling lamB / lamB0
strength sigPN0 / sigPN
stiffness K / K0

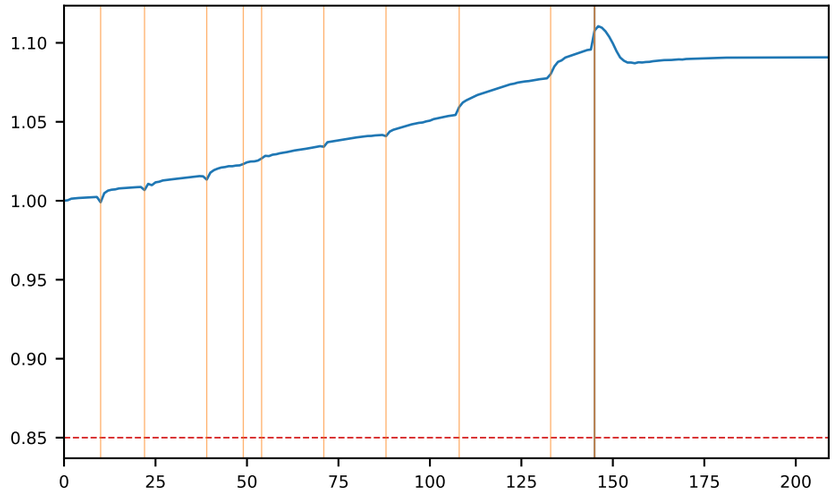

L-bracket: wall thickness (min 1.47, p05 1.55, median 5.3, max 42.2 mm)
as_built
t_min = 1.5 mm
nominal
count of wall samples
wall thickness [mm]
0
500
1000
1500
2000
2500
3000
3500
4000
0
5
10
15
20
25
30
35
40

L-bracket: convergence history

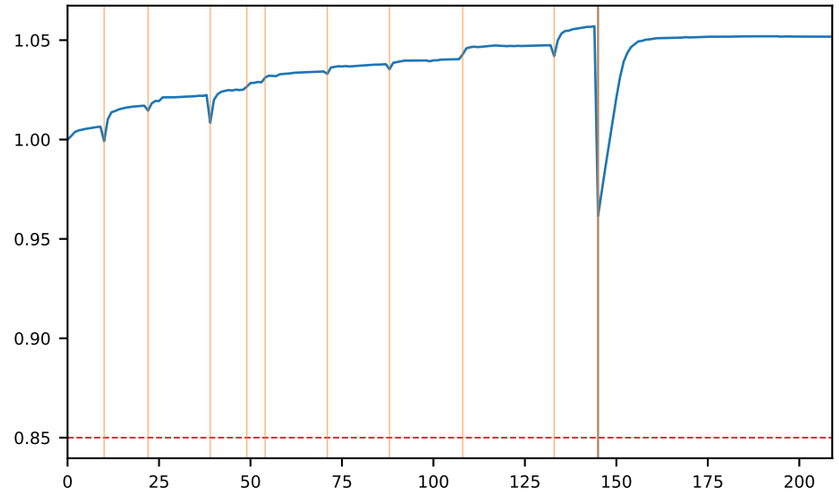
J = weighted objective (accept above ln 1.02)



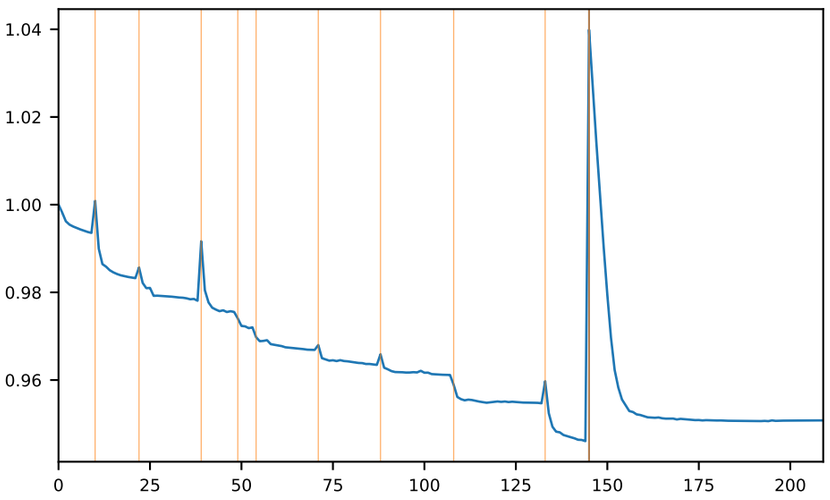
K / K0 (floor 0.85)



P / P0 (floor 0.85)

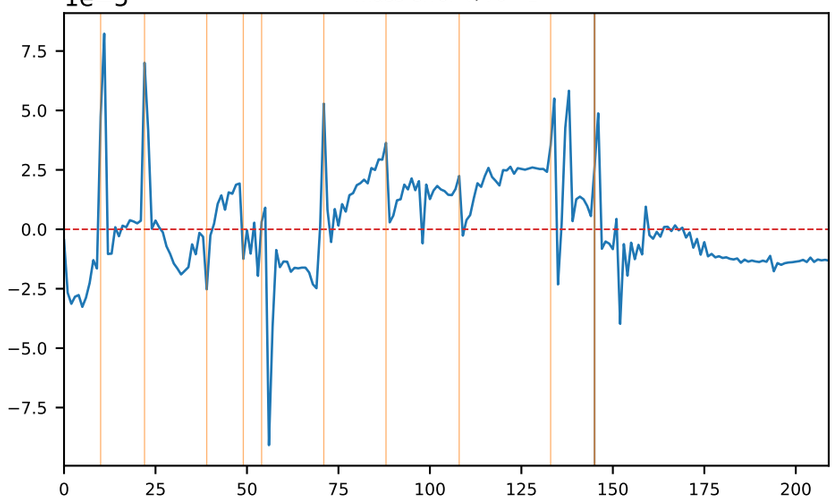


sigPN / sigPN0



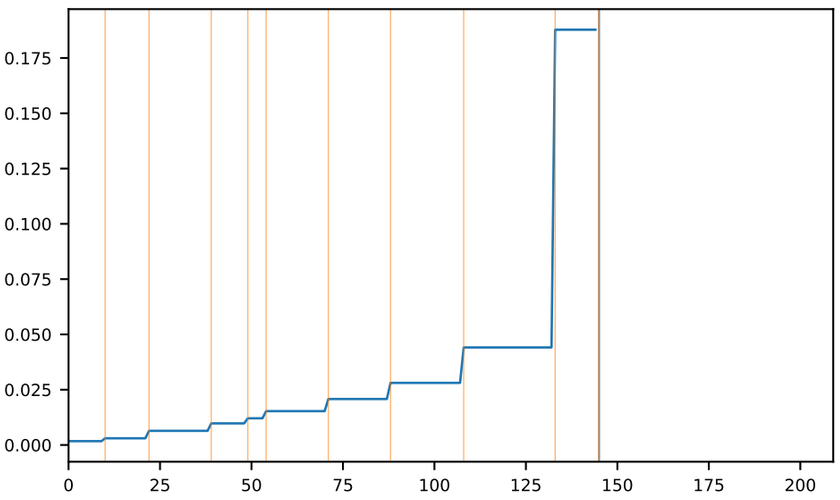
closed fraction

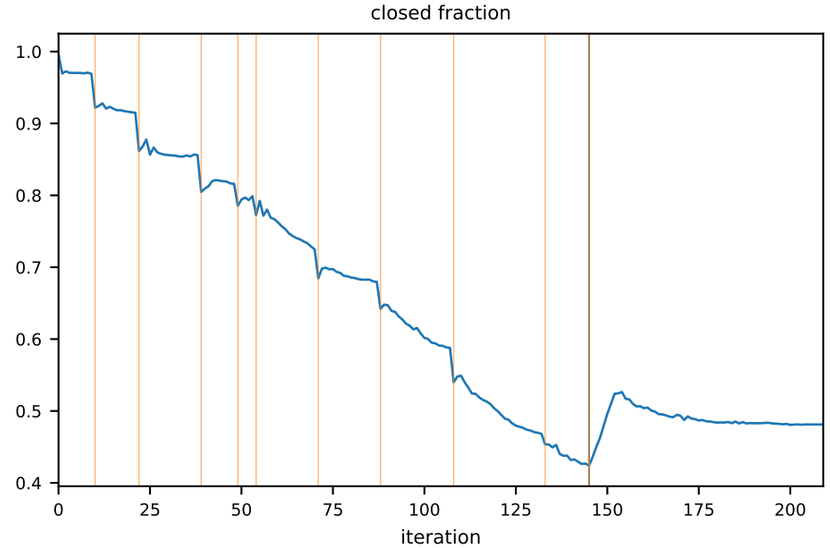
mass error V/V0 - 1

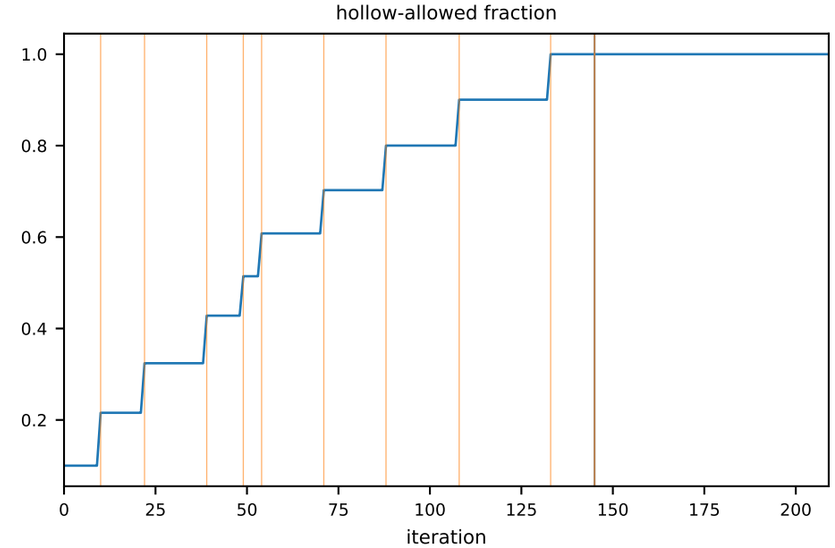


hollow-allowed fraction

R_k (hollow allowed where R <= R_k)



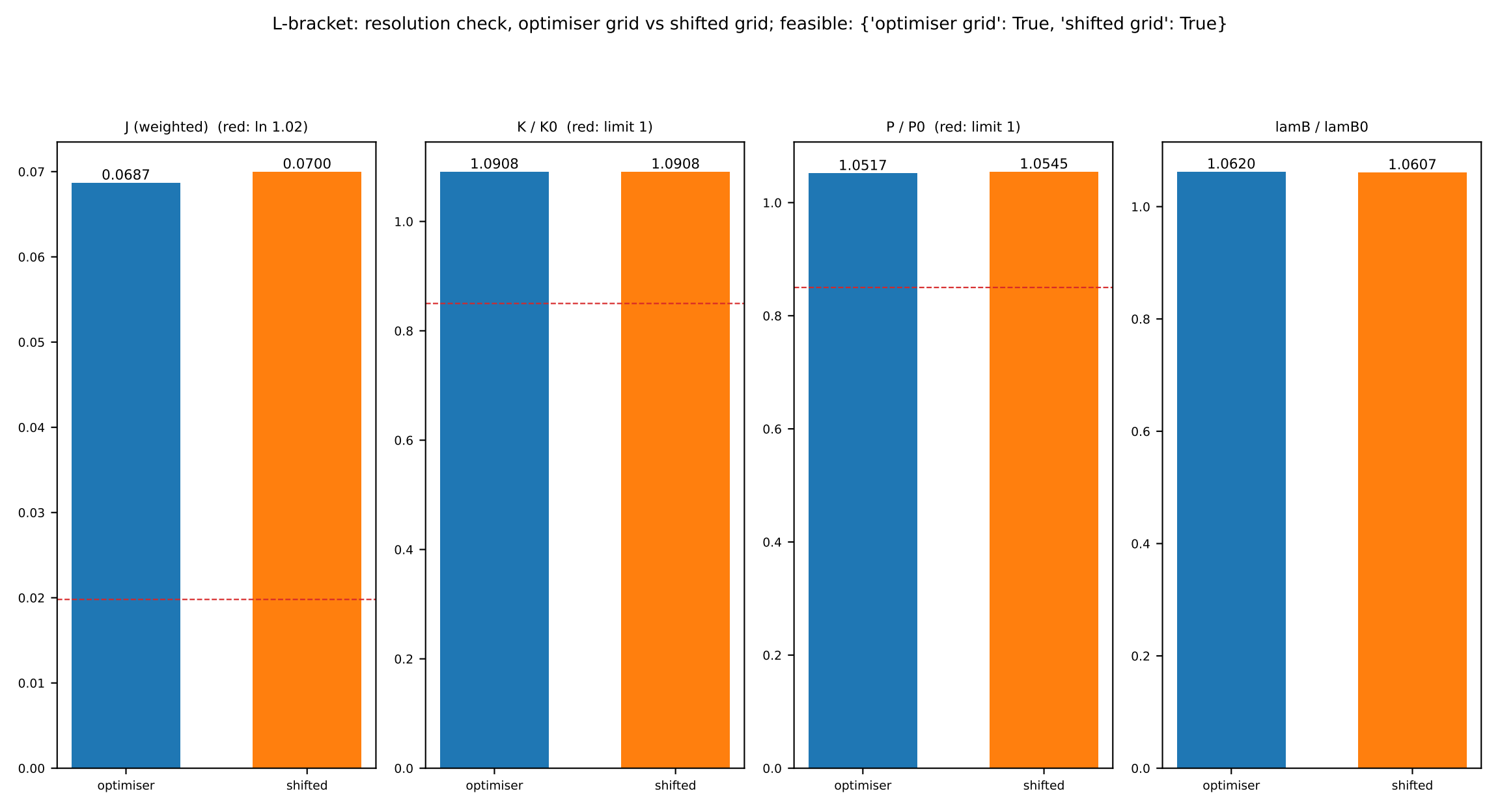


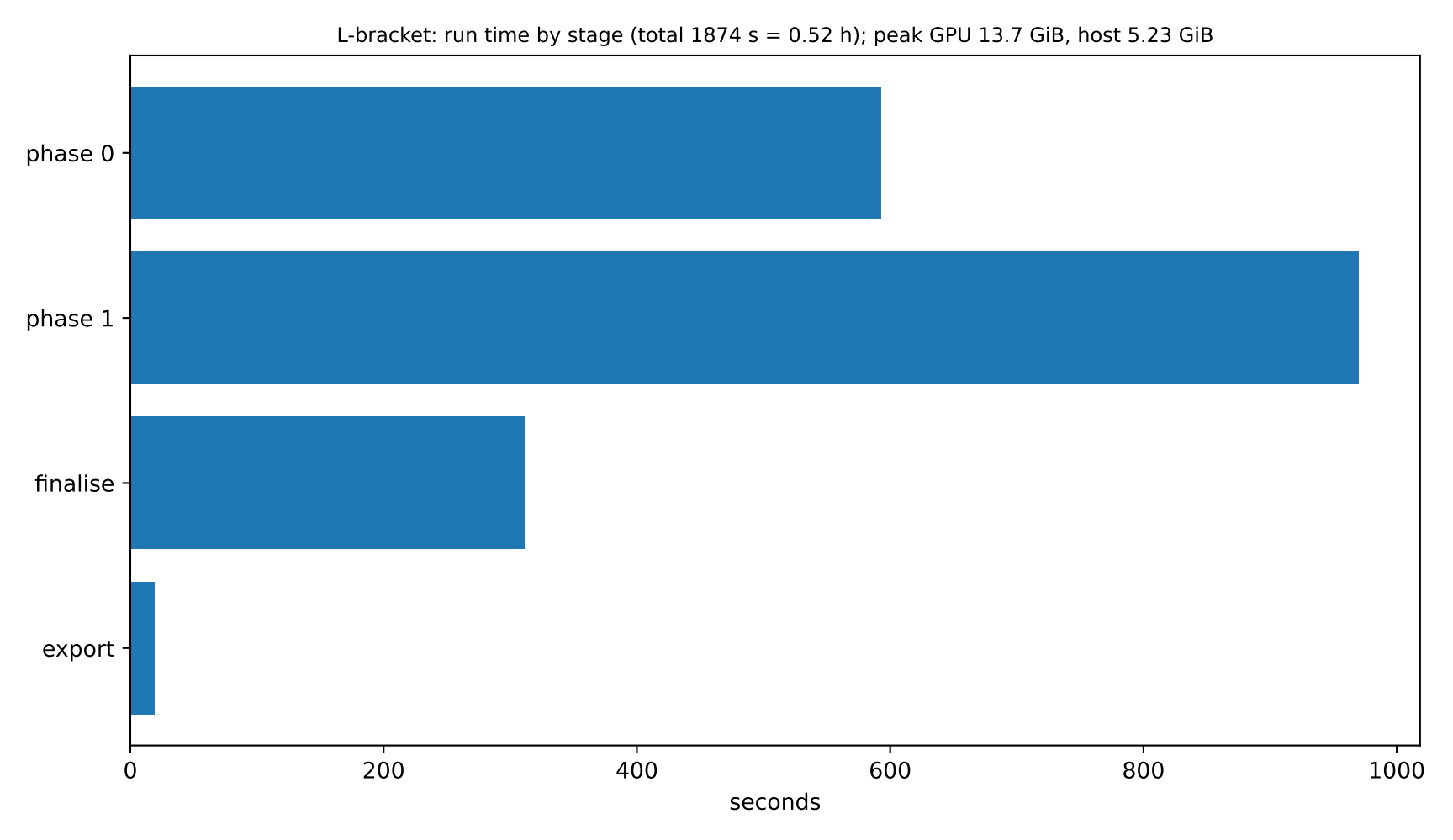


grey vertical: phase boundary
orange vertical: growth step (R_k rise)
iterations 210; growth steps 10
red dashed: constraint / acceptance level

```
enclosed voids: 4;  escape holes: 0;  voids without a hole: 4
  void 0: volume 379.42 mm3, holes 0
  void 1: volume 112.18 mm3, holes 0
  void 2: volume 3977.1 mm3, holes 0
  void 3: volume 34.968 mm3, holes 0

Escape holes sit at the lowest-stress wall points of each void (outward direction in escape_holes.csv).
```


L-bracket: resolution check, optimiser grid vs shifted grid; feasible: {'optimiser grid': True, 'shifted grid': True}
J (weighted) (red: ln 1.02)
0.0687
0.0700
K / K0 (red: limit 1)
1.0908
1.0908
P / P0 (red: limit 1)
1.0517
1.0545
lamB / lamB0
1.0620
1.0607
optimiser
shifted


L-bracket: run time by stage (total 1874 s = 0.52 h); peak GPU 13.7 GiB, host 5.23 GiB
phase 0
phase 1
finalise
export
0
200
400
600
800
1000
seconds